ARSHAD MULLA

Applied AI & NLP | LLMs · Retrieval-Augmented Generation · Efficient Transformer Systems

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SUMMARY

Applied AI Engineer with 1.5+ years of hands-on experience building LLM-powered systems, production-ready RAG pipelines, and NLP applications across research and startup environments.

Experienced in designing and deploying scalable AI systems using LangChain, FAISS, GPT APIs, FastAPI, Docker, and AWS. Strong focus on latency optimization, retrieval grounding, and real-world deployment.

Seeking AI Engineer / NLP Engineer / GenAI roles where I can ship production systems and contribute immediately.

EDUCATION

B.Sc. in Computer Science *Jun 2021 – May 2024*

Royal College of Arts, Science and Commerce — University of Mumbai

EXPERIENCE

Technical Program Analyst — AI & Data | Growfitter (GoToUniversity) *Sep 2025 – Present*

- Built LLM-driven automation pipelines using GPT and Perplexity to manage and enrich a 1,000+ entry global university database, improving data reliability by ~40%.
- Applied prompt engineering and structured retrieval workflows to standardize academic data at scale for an AI-powered web platform.
- Designed scalable knowledge integration techniques to improve retrieval accuracy and system reliability for AI-powered academic search workflows.

AI Researcher (Freelance) | TheSoul Publishing *Aug 2025 – Dec 2025*

- Built NLP classification models to detect AI-generated content, improving language authenticity scores by ~25% across production pipelines.
- Analyzed linguistic patterns and stylistic signals in human-AI co-creation workflows to inform model design and improve detection accuracy.

AI Research Intern | XLEX Batteries *Aug 2025 – Sep 2025*

- Led a team of 4 to design and deploy a domain-specific RAG chatbot using FAISS vector search, embedding models, and LLM re-ranking — reducing internal query resolution time by ~45%.
- Optimized retrieval grounding and response ranking for a structured enterprise knowledge base using LangChain and GPT APIs.

Data Annotation Specialist (Freelance) | Outlier.ai *Jan 2025 – Aug 2025*

- Annotated and validated text, image, and video datasets for NLP and computer vision model training, maintaining high precision and consistency standards.
- Collaborated with data scientists to refine labeling guidelines and QA protocols, improving team throughput and reducing rework rates.

PROJECTS

Generative AI Health Analytics Application

Streamlit · Gemini API · GPT-based reasoning · Python

- Built a multimodal GenAI-powered analytics interface integrating structured reasoning with LLM outputs, emphasizing response reliability and low-latency inference.

Neural Watermarking for Trustworthy LLMs

BART · Hugging Face Transformers · ROUGE · Detection Classifier

- Implemented greenlist-based statistical watermarking for neural abstractive summarization using BART on the CNN/DailyMail dataset.
- Evaluated trade-offs between watermark detectability (F1), generation quality (ROUGE-L), and watermark strength — with findings documented in a research report.

Accelerated NLP Inference: CPU vs MPS vs Mixed Precision *Oct 2025*

PyTorch · BERT · DistilBERT · RoBERTa · Apple MPS · FP16

- Replicated and extended Dr. Forrest Bao's DLBoost vs CUDA BERT study; systematically benchmarked BERT, DistilBERT, and RoBERTa inference across CPU, Apple MPS, and FP16 backends.
- Analyzed latency and throughput trade-offs across batch sizes and sequence lengths; documented practical optimizations for hardware-aware NLP deployment.

AI Image SaaS Platform *Feb 2024 – Apr 2024*

Next.js · MySQL · Cloudinary · Stripe · Shadcn

- Developed an AI-powered image SaaS platform with 95% accuracy in background removal using computer vision and object segmentation; reduced processing time by 30%.
- Integrated Stripe payment gateway and Cloudinary for asset management; premium subscription conversions improved by 15% post-launch.

Robot Navigation System *Aug 2023 – Dec 2023*

Computer Vision · Sensor Fusion · Reinforcement Learning

- Built a robot navigation system using computer vision, sensor fusion, and reinforcement learning — improving path accuracy by 30% and reducing operational latency by 20%.
- Presented at Avishkar 2023 research competition; published in the Journal of Emerging Technologies and Novel Research (JETNR).

TECHNICAL SKILLS

LLM & GenAI: GPT APIs, Gemini API, Hugging Face Transformers, LangChain, LangGraph
RAG & Retrieval: FAISS, Pinecone, Sentence-Transformers, Embedding Models, Prompt Engineering
ML & NLP: PyTorch, TensorFlow, BERT, RoBERTa, BART, Model Evaluation (ROUGE, F1), Scikit-learn, NumPy, Pandas
Deployment: FastAPI, Docker, AWS EC2, REST APIs
Programming: Python, SQL, JavaScript

PUBLICATIONS

Robot Navigation using Reinforcement Learning and Sensor Fusion
Journal of Emerging Technologies and Novel Research (JETNR), Dec 2023 — Research Presenter, Avishkar 2023

CERTIFICATIONS

- Python for Machine Learning — Great Learning (Oct 2023)
- Introduction to Deep Learning — Great Learning (Apr 2023)